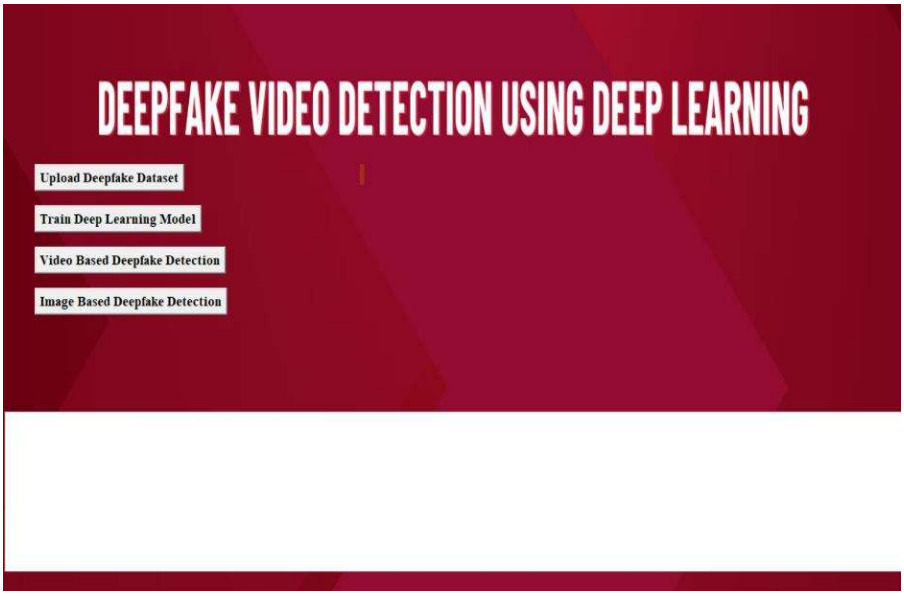
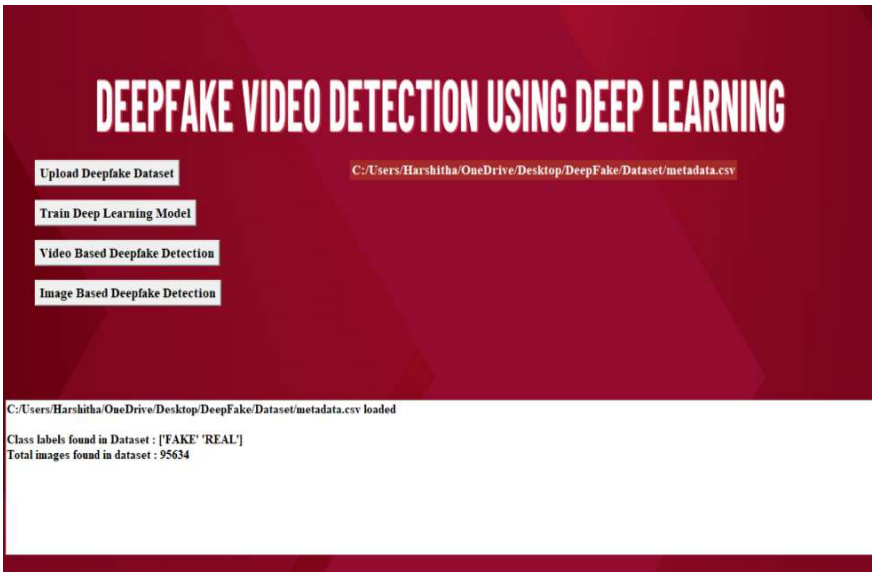
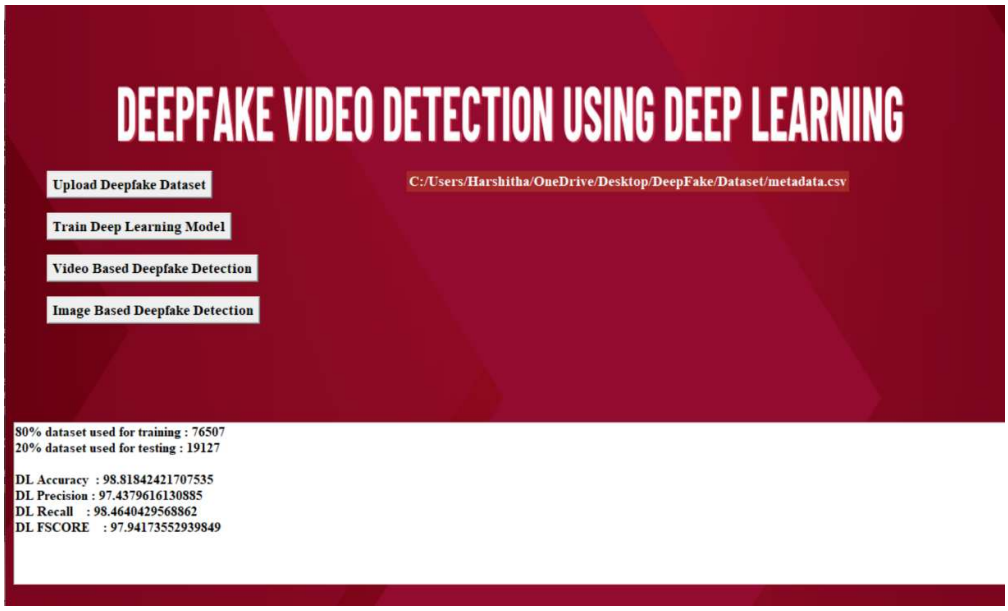
To run project double click on run.bat file to get below screen



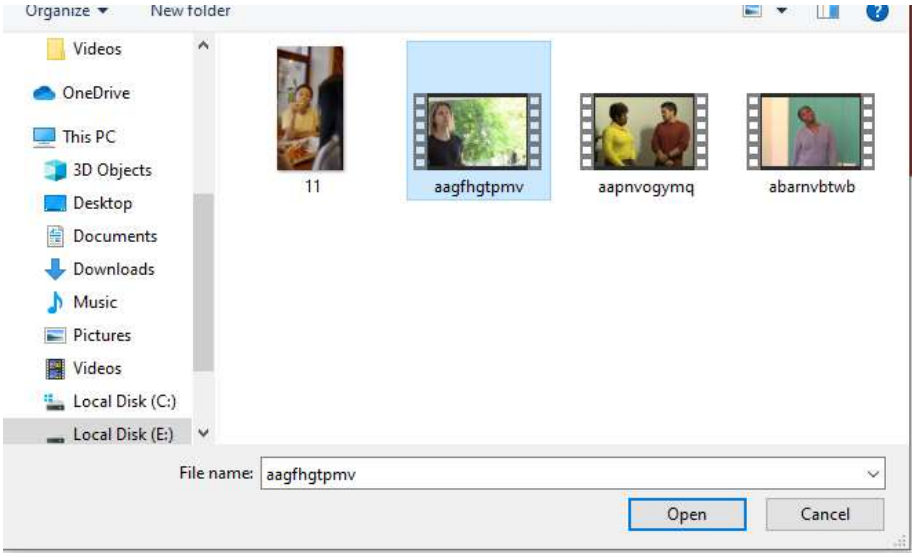
In above screen click on 'Upload Deepfake Faces Dataset' button to load dataset and get below page



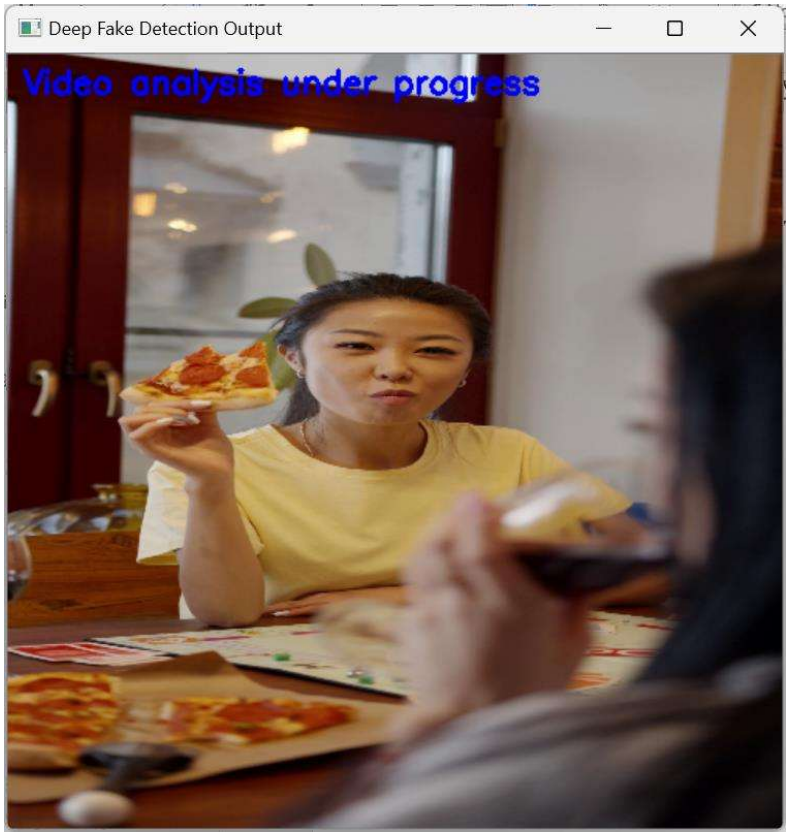
In above screen can see dataset loaded and can see different class labels found in dataset and then can see number of images found in dataset and now click on 'Train DL Model' button to train algorithm and get below page



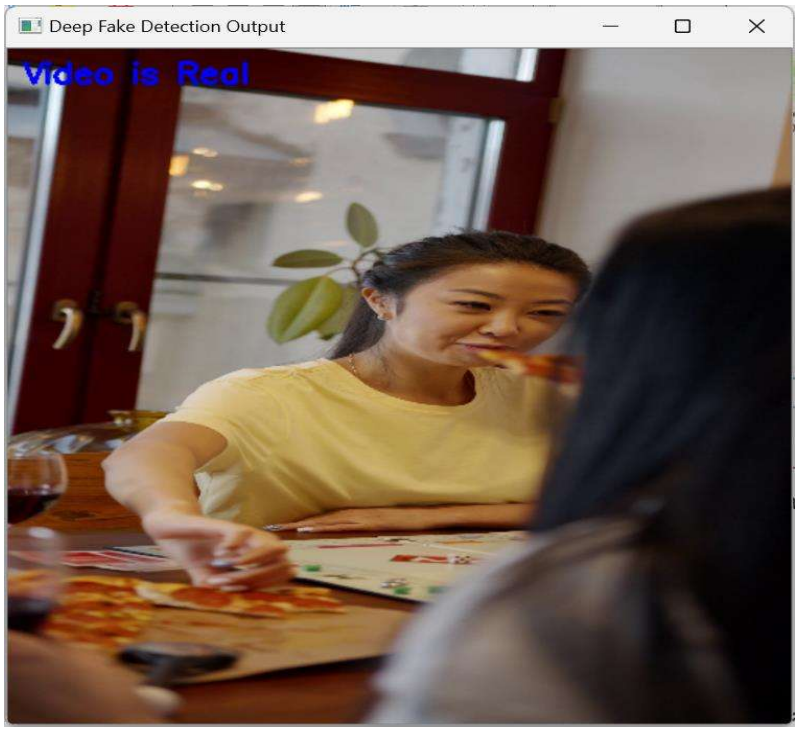
In above screen can see train and test dataset size and then can see DL got 99% accuracy and can see other metrics like precision, recall and FSCORE. Now click on 'Video Based Deepfake Detection' button to upload test video and get below page



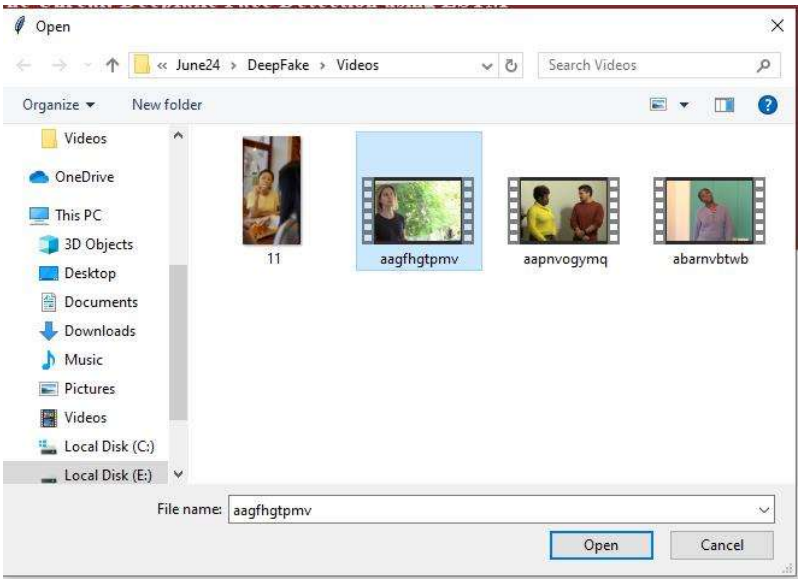
In above screen selecting and uploading 11.mp4 video and then click on 'Open' button to start analysing video and get below page



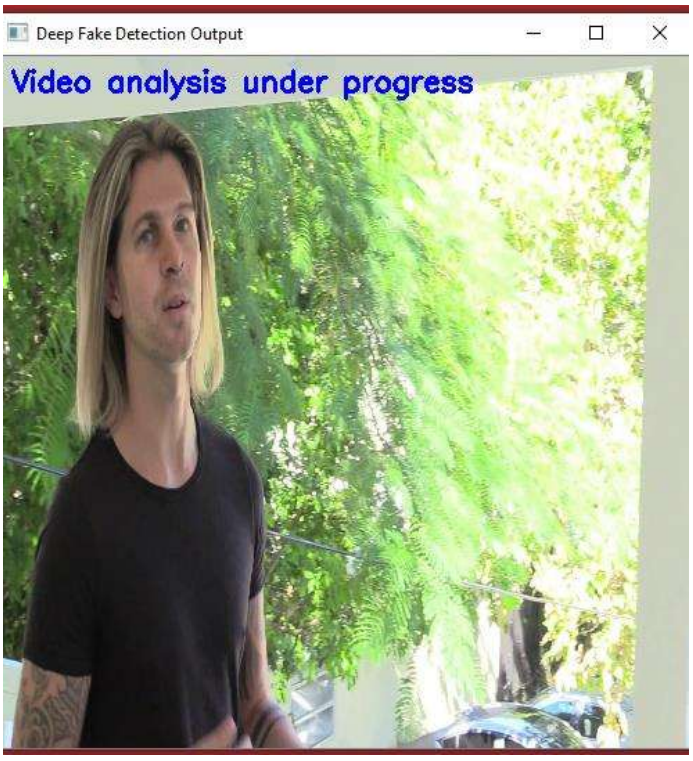
In above screen in blue colour text can see video analysis started and after thorough analysis by DL will get below output



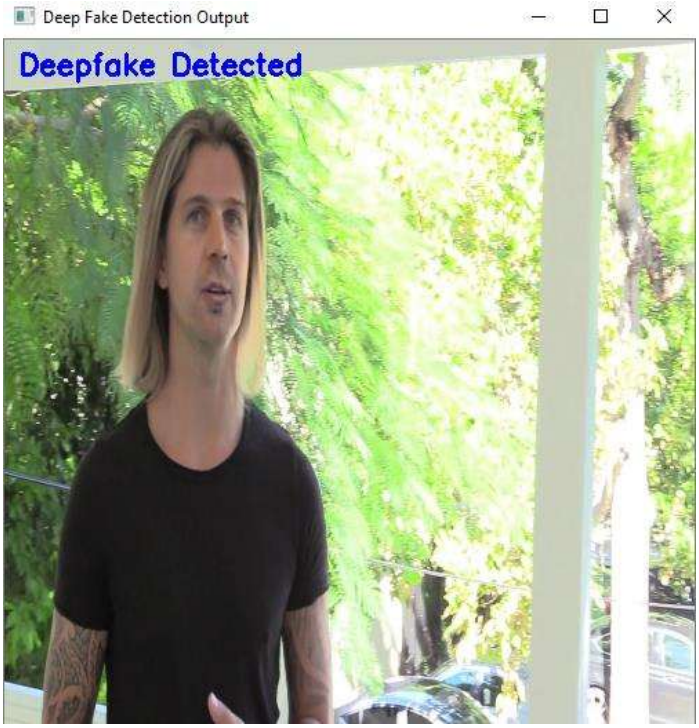
In above screen uploaded video predicted as 'Real' and now test with other fake video



In above screen uploading 'aagfhgtpmv.mp4' video and then click on 'Open' button to load video and get below output



In above screen video analysis under progress and once after analysis will get below output



In above screen uploaded video is detected as Deepfake and similarly you can upload and test other video

4.6 Conclusion for Deepfake Detection

Deep learning techniques have proven to be highly effective in detecting deep fakes by identifying subtle artifacts and inconsistencies in manipulated media. Models such as Convolutional Neural Networks (CNNs) and Transformer-based architectures have shown impressive accuracy in distinguishing real content from forgeries. However, the performance of these models depends heavily on the availability of large and diverse datasets that represent real-world scenarios. One major challenge is the generalization of detection methods across different datasets and unseen deep fake generation techniques, which highlights the need for continuous research and development. Additionally, as deep fake generation methods become more sophisticated, detection systems must evolve to keep pace. Ethical considerations, such as privacy and potential misuse, also play a critical role in the development and deployment of these technologies. Overall, while deep learning provides a robust foundation for deep fake detection, sustained efforts are required to address evolving challenges and ensure its effectiveness in real-world applications.